

Module: ADS1 Algorithms and Data Structure 1
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Type: Tutorial

TD 04: Arrays

Exercise 1

Write an algorithm or C program that reads n integer values into an array and displays them in reverse order.

Exercise 2

Write an algorithm or C program to read two arrays of the same size and compute a third array where
 $C[i] = A[i] + B[i]$.

Exercise 3

Write an algorithm or C program to check if a given array is sorted in ascending order.

Exercise 4

Write an algorithm or C program to merge two arrays of the same size, both sorted in descending order, into a new descending-sorted array.

Exercise 5

Write an algorithm or a C program that rearranges the elements of a 1D array so that all **odd** numbers appear first, followed by all **even** numbers. The relative order of the numbers does not need to be preserved.

After rearranging the array, display the final result.

Exercise 6

Write an algorithm or C program to find a pair of elements in an array whose sum equals a given value S .

Exercise 7

Write an algorithm or C program to compute and display the sum of each row of a matrix.

Exercise 8

Write an algorithm or C program to find the maximum element in a matrix and display its value along with its position (row and column).

Exercise 9

Write an algorithm or C program to compute the transpose of a 2D array (matrix).

Exercise 10

Write an algorithm or C program to multiply two matrices (2D arrays).